Shams Sajid Rahman

+880-1325075532 | shamssajid121@gmail.com | linkedin.com/in/shams-sajid-rahman | github.com/Shams-Sajid-Rahman
Dhaka Cantonment, Dhaka-1206, Bangladesh

Professional Summary

Computer Science and Engineering graduate from North South University with a strong foundation in full-stack software engineering, machine learning, deep learning, and natural language processing. Experienced in designing and implementing AI-powered applications, multi-model regression systems, scalable fronted and backend APIs. Currently contributing to active research text-to-medical video generation, with a publication in progress. Recognized for technical leadership and community involvement through the NSU ACM Student Chapter.

Technical Skills

Languages: C, C++, Java, JavaScript, Dart, SQL, HTML5, CSS3, Python

ML & AI: PyTorch, TensorFlow, Scikit-learn, OpenCV, Hugging Face (Transformers & Diffusers), PEFT & LoRA, LangChain, Google Gemini API, CBAM, MC-Dropout

Backend & Frontend: React.js, Node.js, Express.js, EJS, Flutter (Dart), RESTful API Design, MVC Architecture

Databases: PostgreSQL, MySQL, Firebase Realtime Database, MongoDB

Developer Tools: Git, GitHub, Jupyter Notebook, Google Colab, VS Code, PyCharm

Data Science: Pandas, NumPy, Matplotlib, Seaborn, GridSearchCV, Cross-Validation, Model Evaluation Metrics

Education

North South University — Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering – CGPA: 3.35 / 4.00 — *January 2022 – May 2026*

Work Experience

FlyRank AI — Remote
Machine Learning Intern — *July 2026 – September 2026*

- Completed a structured Machine Learning Internship Program at FlyRank Corp., applying practical ML skills through assignments and a capstone project that was reviewed and accepted by the team.
- Demonstrated excellence in technical competency, professional conduct, and collaborative contribution to the FlyRank AI team. Certificate ID: FR-D11-AF08E-4D5DE.

Projects

CRUD-Based Job Portal | *Node.js, Express.js, PostgreSQL, EJS* | GitHub — 2023

- Delivered a full-stack job portal with complete CRUD operations, session-based authentication, and role-based access control (RBAC) following MVC architecture and RESTful API design principles.
- Architected a normalized PostgreSQL schema and Express.js API with modular middleware, supporting dual-role workflows for job seekers and employers with clean separation of concerns.

Customer Management API | *Node.js, Express.js, Prisma, MySQL, Docker, Jest* | GitHub — 2026

- Designed a strictly layered REST API (routes → controllers → services → repositories) in Node.js/Express with Prisma and MySQL, managing customers and their nested phone number, address, and document sub-resources with cascade deletes and transactional updates.
- Wrote 69 automated Jest/Supertest tests across 5 suites (DB-independent, CI-safe) covering validation, pagination, and error handling; containerized the stack with Docker Compose and documented all endpoints via Swagger/OpenAPI.

RBAC Admin API | *Node.js, TypeScript, Express 5, PostgreSQL, Prisma, JWT* | GitHub — 2026

- Built a role-based access control (RBAC) backend for an e-commerce admin panel using Node.js, TypeScript, Express 5, and PostgreSQL via Prisma, covering 9 modules (auth, users, roles, permissions, media, categories, brands, attributes, product variants) across 55+ REST endpoints.
- Implemented JWT authentication with access/refresh token rotation, server-side revocation, and rate-limited login to prevent brute-force attacks; wrote an automated audit script that verified permission enforcement across all 55 routes with a 100% pass rate.

Research Experience

Accuracy–Robustness Tradeoff in Multi-Source Alzheimer's Disease Classification 2025
North South University *Dhaka, Bangladesh*

- Benchmarked 5 deep learning architectures (ResNet-18, MobileNet-V3, NeuroFusion, DeepCNN-SE-CBAM, SKTripletFusion) across 89,955 MRI images from 3 independent datasets (Kaggle, OASIS, ADNI) to quantify accuracy-robustness tradeoffs under cross-source domain shift.
- Demonstrated that DeepCNN-SE-CBAM with dual-attention mechanisms achieves 91% real-world robustness, outperforming 99%+ lab-accuracy models that collapsed under clinical noise conditions.

MedT2V: Text-to-Medical Video Generation Framework with PME and DAF 2026
North South University *Dhaka, Bangladesh*

- Constructed an end-to-end text-to-video pipeline trained on 55,000 medical video-caption pairs using LoRA fine-tuning of the Wan2.1-T2V-1.3B diffusion model.
- Engineered Pyramidal Pseudo-Mamba Encoding (PME) and Direction-Aware Fusion (DAF) for multi-scale temporal modeling and enhanced text-video semantic alignment.

Publications

Accuracy-Robustness Tradeoff in Multi-Source Alzheimer's Disease Classification.
Published in **IEEE QPAIN 2026**, Chittagong, Bangladesh. IEEE Xplore. DOI: 10.1109/QPAIN69676.2026.11546348

Leadership & Achievements

NSU ACM Student Chapter Dhaka, Bangladesh
General Secretary *2026 – Present*

- Organized 15+ programming contests, technical workshops, and knowledge-sharing sessions for a 300+ member community, sustaining consistent event engagement over 4 years of chapter involvement.
- Earned the Best Member Award, being selected from over 300 active members, as well as for 4 years of outstanding technical contributions and sustained leadership across all chapter events.

Complete Web Development Course with Programming Hero Online
Batch 12 | Instructor: Jhankar Mahbub *June 2025 – Present*

- Currently pursuing an intensive full-stack web development bootcamp covering modern frontend and backend technologies including React.js, Node.js, and MongoDB.

Professional WordPress Mastery Zero to Hero Online
Certificate of Achievement – UY LAB *Feb 2025 – May 2025*

- Completed an intensive 3-month professional training program covering end-to-end WordPress development, from fundamentals to advanced mastery. Issued May 20, 2025 (ID: WMZH-02502027839).